

Far-East Scarlet-Like Fever (FESLF): A Comprehensive Disease Characterization

Disease: Far-East Scarlet-Like Fever **MONDO ID:** MONDO:0041536 **Category:** Infectious Disease **Report type:** Multi-iteration autonomous literature synthesis for disease knowledge-base population

Summary

Far-East Scarlet-Like Fever (FESLF) is the severe, systemic, toxin-mediated clinical-epidemic form of infection by the enteric Gram-negative bacterium *Yersinia pseudotuberculosis*, occurring predominantly in the Russian Far East, Siberia, and Japan. It is *not* a genetic or heritable disorder; it is an infectious disease whose distinctive severity is driven by strain-specific virulence factors carried by the "Asian clade" of *Y. pseudotuberculosis*. Where *Y. pseudotuberculosis* in Europe typically causes a self-limiting gastroenteritis or mesenteric adenitis, the Far-Eastern strains produce a scarlet-fever/Kawasaki-disease-like multisystem illness characterized by high fever, scarlatiniform rash, desquamation, strawberry tongue, cracked lips, conjunctivitis, gastrointestinal and hepatic involvement, and arthralgias.

The central molecular determinant of FESLF is the horizontally acquired superantigen gene **ypmA**, which encodes the **Y. pseudotuberculosis-derived mitogen A (YPMa)**. This structurally unique superantigen (jelly-roll fold, no homology to staphylococcal/streptococcal superantigens) cross-links MHC class II (HLA-DR) with T-cell receptor V β elements (V β 3, 9, 13.1, 13.2), triggering a polyclonal, V β -restricted T-cell activation and a systemic IFN- γ /TNF- α cytokine storm. Far-Eastern FESLF strains additionally carry the **pVM82 (82-MDa) plasmid**, a molecular marker present *only* in strains causing the clinical-epidemic FESLF manifestation. The severity of disease is proportional to the intensity of TCR-V β engagement, as demonstrated by point-mutation studies in which weakened V β binding proportionally reduced T-cell overstimulation, cytotoxicity, and cytokine production.

FESLF sits at the intersection of scarlet fever, Japanese "Izumi fever," and Kawasaki disease (KD). *Y. pseudotuberculosis*-associated KD carries a significantly elevated risk of coronary artery lesions and IVIG resistance. The characteristic target lesion of the underlying infection is

a granulomatous terminal ileitis and mesenteric lymphadenitis that histologically mimics Crohn's disease. Post-infectious immune sequelae — HLA-B27-associated reactive arthritis and erythema nodosum — are major contributors to long-term morbidity. Diagnosis rests on stool/blood culture, serology (agglutinating antibodies and anti-YPM antibody titers), and PCR for virulence genes; treatment is antibiotics plus supportive care, with IVIG/aspirin in KD-presenting pediatric cases. Prevention is through food and water hygiene, as the organism is transmitted through contaminated produce (notably carrots) and water.

1. Disease Information

Overview. FESLF is a severe systemic inflammatory disease caused by *Yersinia pseudotuberculosis*. It represents the special "clinical-epidemic" manifestation of pseudotuberculosis that occurs sporadically and in outbreaks in Russia and Japan, in contrast to the milder self-limiting gastroenteritis seen in Europe ([PMID: 26819960](#)). A case series of 12 culture-confirmed children established the multisystem phenotype and noted that the illness "resembled those of Izumi fever, an illness that occurs epidemically in Japan" ([PMID: 6344044](#)).

Key identifiers.

Resource	Identifier
MONDO	MONDO:0041536
Causative organism (NCBI Taxonomy)	<i>Yersinia pseudotuberculosis</i> , txid633
MeSH	Yersinia pseudotuberculosis Infections (D015008)
ICD-10	A28.2 (Extraintestinal yersiniosis) / A04.8
OMIM	Not applicable (non-genetic, infectious disease)
Orphanet	Not a listed rare Mendelian disease

Synonyms / alternative names. Far East scarlet-like fever; Far Eastern scarlet-like fever; Izumi fever (Japan); Far-Eastern scarlatiniform fever; a clinical-epidemic form of pseudotuberculosis / extraintestinal yersiniosis. The relation to Izumi fever and Kawasaki disease is explicitly established in the literature ([PMID: 6344044](#); [PMID: 39780644](#)).

Source of information. The knowledge base is derived from **aggregated disease-level resources** — case series, outbreak investigations, genomic/phylogenetic studies, and mechanistic in vitro/animal experiments — rather than individual EHR-derived patient records.

2. Etiology

Disease causal factors. FESLF is caused by **infection** with *Y. pseudotuberculosis*, specifically Far-Eastern "Asian clade" strains carrying the superantigen gene *ypmA* and (in clinical-epidemic strains) the pVM82 plasmid (PMID: 26819960; PMID: 30695393; PMID: 39780644). The disease is fundamentally toxin-mediated: the YPM superantigen drives the systemic manifestations.

Environmental risk factors. Transmission is foodborne/waterborne; documented vehicles include contaminated grated carrots (PMID: 23852698) and other fresh produce and water. The organism is psychrotrophic (grows at refrigeration temperatures), enabling amplification in stored vegetables. Age is a risk modifier — the underlying infection disproportionately affects children, and *Yersinia*-associated KD occurs at older onset age (3.05 ± 2.20 y vs 2.31 ± 2.05 y; $p = 0.03$) (PMID: 17129979). Immunocompromise and iron overload predispose to invasive/septicemic disease (PMID: 42448289).

Genetic (host) risk factors. Host genetics do not cause FESLF but modify post-infectious complications: **HLA-B27** strongly predisposes to reactive arthritis after *Y. pseudotuberculosis* infection (PMID: 23852698; PMID: 12922960). The TCR V β repertoire of the host (V β 3/9/13.1/13.2) determines which T cells respond to YPM. NOD2 and ATG16L1 autophagy polymorphisms have been implicated in susceptibility to *Yersinia* mucosal disease in the IBD context (PMID: 42661478).

Protective factors. No germline protective alleles are established for FESLF itself. Antibiotic therapy and prior immunity (anti-YPM antibodies) modulate course. Food-hygiene behaviors are the principal protective/preventive measures.

Gene–environment interactions. The defining interaction is between the bacterial superantigen (environmental/infectious agent) and the host MHC-II/TCR-V β genotype: the same toxin produces variable illness depending on host HLA-DR and V β repertoire. Separately, HLA-B27 $\times$ *Yersinia* infection markedly raises reactive-arthritis risk — a classic gene–environment interaction.

3. Phenotypes

The core phenotype was quantified in a case series of 12 children with culture-confirmed *Y. pseudotuberculosis* (stool culture plus ≥ 4 -fold agglutinating-antibody rise); clinical findings present in $\geq 50\%$ of patients are listed below ([PMID: 6344044](#)).

Phenotype	Type	Frequency ($\geq 50\%$ cohort)	Suggested HPO term
Fever	Symptom	$\geq 50\%$	HP:0001945
Scarlatiniform rash	Skin sign	$\geq 50\%$	HP:0000988 (Skin rash)
Diarrhea	Symptom	$\geq 50\%$	HP:0002014
Desquamation	Skin sign	$\geq 50\%$	HP:0007556 (Palmoplantar desquamation)
Strawberry tongue	Sign	$\geq 50\%$	HP:0000206 (Glossitis, related)
Vomiting	Symptom	$\geq 50\%$	HP:0002013
Red, cracked lips	Sign	$\geq 50\%$	HP:0000202 (Oral cavity abnormality)
Abdominal pain	Symptom	$\geq 50\%$	HP:0002027
Arthralgias	Symptom	$\geq 50\%$	HP:0002829
Hepatomegaly	Sign	$\geq 50\%$	HP:0002240
Conjunctivitis	Sign	$\geq 50\%$	HP:0000509

Direct quote: "Clinical findings in 50% or more of patients were fever, rash, diarrhea, desquamation, strawberry tongue, vomiting, red and cracked lips, abdominal pain, arthralgias, hepatomegaly and conjunctivitis" ([PMID: 6344044](#)).

Additional phenotypes documented in adults/severe cases: membranous fingertip/interdigital desquamation ([PMID: 42778384](#); [PMID: 18411766](#)), hepatic dysfunction and elevated inflammatory markers, erythema nodosum (42% of children in one carrot-borne outbreak), and reactive arthritis. Laboratory abnormalities include leukocytosis, elevated CRP/ferritin, and in severe cases thrombocytopenia with DIC ([PMID: 16366361](#)).

Onset/severity/progression. Onset is acute (childhood-predominant). Severity ranges mild to severe; most cases are self-limited over days to weeks, but a subset progresses to systemic toxicity, coronary involvement (KD-like), DIC, or septicemia. Reactive arthritis can become chronic (>6 months) and rarely fatal (secondary amyloidosis).

Quality-of-life impact. Acute illness impairs feeding, activity, and school attendance. Chronic reactive arthritis and ankylosing spondylitis produce sustained functional disability; fatal amyloidosis with uremia has been reported in long-term follow-up ([PMID: 7554560](#)).

4. Genetic / Molecular Information

FESLF is not a human genetic disease — there are no causal human genes, pathogenic germline variants, or chromosomal abnormalities. The relevant "genetics" are those of the pathogen and of host susceptibility loci.

Pathogen virulence genes.

Determinant	Function	Association with FESLF
ypmA (encodes YPMa)	Superantigen mitogen	Present in 96.2% of Far-Eastern strains; hallmark of systemic disease (PMID: 17163133)
ypmB, ypmC	Superantigen variants	ypmB Far-East-restricted cluster B; ypmC rare (PMID: 21131531)
pVM82 (82-MDa) plasmid	Uncharacterized virulence	Present <i>only</i> in clinical-epidemic FESLF strains (PMID: 30695393)
pYV virulence plasmid (virF)	Type III secretion system / Yops	Universal in pathogenic strains (PMID: 18242014)
inv (invasin)	β 1-integrin adhesion, M-cell invasion	Universal; mediates epithelial crossing (PMID: 25576025)
HPI / irp2	High-pathogenicity island (iron uptake)	Absent in dominant FESLF genogroup (PMID: 17163133)
pil (type IV pilus)	Adhesion	Co-acquired with ypm horizontally (PMID: 15784605)

The dominant systemic-infection genogroup is **pYV⁺ / ypmA⁺ / HPI⁻** (95.8% of Siberian/Far-Eastern strains) ([PMID: 17163133](#)). The ypm and pil genes were laterally (horizontally) acquired and are significantly co-associated ([PMID: 15784605](#)).

Host susceptibility loci. HLA-B27 (reactive arthritis modifier); HLA-DR (YPM presentation); TCR V β genes BV3S1, BV9, BV13 (target of YPM); C4B copy number (modifies Yersinia-microbiota interaction in pediatric IBD) ([PMID: 28832994](#)).

Epigenetics. No disease-specific human epigenetic signature is established for FESLF. Superantigen-driven T-cell activation induces transcriptional/epigenetic reprogramming toward effector/memory phenotypes ([PMID: 11937534](#)), but this is not disease-defining.

5. Environmental Information

Infectious agent. *Yersinia pseudotuberculosis* (family Yersiniaceae; NCBI Taxonomy txid633), a Gram-negative, psychrotrophic, facultatively anaerobic coccobacillus. Serotypes O:1 (incl. Ib) and O:3 are the principal causes of documented outbreaks and post-infectious complications; serotypes 4b and 1b predominate in fatal Japanese monkey outbreaks ([PMID: 18242014](#)).

Environmental factors. The organism is widespread in the terrestrial environment and can persist in soil and water; in wildlife reservoirs (rodents, wild boars, birds), and even the marine habitat ([PMID: 42431144](#); [PMID: 29980552](#)). Its cold tolerance allows growth in refrigerated foods.

Lifestyle factors. Consumption of raw or improperly washed vegetables (carrots, lettuce) and untreated water; contact with animal reservoirs. Outbreaks are typically food/water-associated. The dominant environmental "cause" is ingestion of contaminated food, then M-cell invasion in the terminal ileum.

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Ingestion** of *Y. pseudotuberculosis* (Asian-clade, ypmA⁺, pVM82⁺) in contaminated food/water → **leads to** delivery of viable bacteria to the terminal ileum.
2. Bacterial **invasin binds host β 1-integrins on M cells** of Peyer's patches → **results in** translocation across the intestinal epithelium ([PMID: 25576025](#)).
3. Bacteria reach the **mesenteric lymph nodes, liver, and spleen**; the type III secretion system (pYV/Yops) is selectively translocated into **professional phagocytes** (neutrophils, macrophages, dendritic cells), disarming innate clearance → **leads to** local persistence and **granulomatous terminal ileitis / mesenteric lymphadenitis** ([PMID: 20148898](#); [PMID: 34628159](#)).
4. The bacterium **secretes/expresses the superantigen YPMa** → which **binds MHC class II (HLA-DR) on antigen-presenting cells** outside the conventional peptide groove ([PMID: 10406939](#)).
5. The MHC-II–YPM complex **cross-links T-cell receptors bearing V β 3, 9, 13.1, 13.2** → **results in** massive polyclonal (V β -restricted) T-cell activation ([PMID: 9287138](#); [PMID: 17369701](#)).

6. Activated V β ⁺ T cells **migrate from blood to liver within 1 h** and **release a cytokine storm** (IFN- γ by 4 h, plus TNF- α , IL-1, IL-6, IFN- α) → **leads to** systemic inflammation, hepatic injury, and toxic shock (PMID: 15003813; PMID: 12449699).
7. Systemic cytokine excess and vascular inflammation **produce the clinical syndrome**: fever, scarlatiniform rash, desquamation, strawberry tongue, conjunctivitis, hepatomegaly → the FESLF/Kawasaki-like phenotype.
8. **Branch A (severity)**: the *intensity* of TCR-V β binding scales the magnitude of T-cell overstimulation, cytotoxicity, and cytokine output → **determines** disease severity (PMID: 10087177).
9. **Branch B (cardiac)**: in a subset, superantigen-driven vasculitis → **coronary artery dilation/aneurysm** (Kawasaki-disease phenotype), with higher coronary-lesion rates and IVIG resistance in Yersinia-associated KD (PMID: 17129979).
10. **Branch C (post-infectious autoimmunity)**: in HLA-B27⁺ hosts, molecular/immune cross-reactivity → **reactive arthritis, ankylosing spondylitis, erythema nodosum**, rarely secondary amyloidosis (PMID: 23852698; PMID: 7554560).

Detail by category

- **Molecular pathways**: TCR/CD3 → ZAP-70 → PLC γ /PKC/NF- κ B and Ras-MAPK signaling driving IL-2/IFN- γ transcription (superantigen-driven); β 1-integrin → PI3K signaling in invasin-mediated neutrophil activation and NET formation (PMID: 25576025); RhoG/Rac1 GTPase modulation by invasin/Yops during invasion (PMID: 19208761).
- **Cellular processes**: massive T-cell proliferation and later anergy/deletion; macrophage/DC activation; neutrophil extracellular trap (NET) release; inflammasome activation and pyroptosis in Yersinia mucosal disease (PMID: 42661478); granuloma formation.
- **Protein dysfunction (toxin structure)**: YPMa is a ~150-aa protein adopting a **jelly-roll fold** resembling viral capsid/TNF-superfamily proteins, with no sequence homology to other superantigens; crystallized in space group C2, 1.8 Å resolution; an internal disulfide (S–S) bond is essential for activity (PMID: 12832802; PMID: 17369701; PMID: 10406939).
- **Immune system involvement**: superantigen-mediated polyclonal T-cell activation, IFN- γ -dominated cytokine storm; T-cell-dependent toxicity (toxic shock in BALB/c but **not** T-cell-deficient SCID mice) (PMID: 15003813).
- **Tissue damage mechanisms**: IFN- γ /TNF- α -mediated hepatocellular injury; immune-complex and vasculitic damage to coronary vessels; granulomatous necrosis in ileum/lymph nodes.

- **Suggested GO terms:** GO:0042110 (T cell activation), GO:0032609 (interferon-gamma production), GO:0050852 (T cell receptor signaling pathway), GO:0006955 (immune response), GO:0002250 (adaptive immune response), GO:0001816 (cytokine production).
 - **Suggested CL terms:** CL:0000624 (CD4⁺ T cell), CL:0000625 (CD8⁺ T cell), CL:0000775 (neutrophil), CL:0000235 (macrophage), CL:0000451 (dendritic cell), CL:0000794 (CD8⁺ cytotoxic T cell).
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7. Anatomical Structures Affected

Organ level. - **Primary:** terminal ileum (UBERON:0002116), mesenteric lymph nodes (UBERON:0002509), liver (UBERON:0002107). - **Secondary/systemic:** skin (UBERON:0002097), oral mucosa/tongue (UBERON:0001723), conjunctiva (UBERON:0001811), coronary arteries (UBERON:0001621) in KD-like cases, joints/synovium (UBERON:0002217) in reactive arthritis, spleen (UBERON:0002106), pancreas (UBERON:0001264; *Yersinia* pancreatitis, [PMID: 22416431](#)). - **Body systems:** digestive, lymphatic/immune, integumentary, cardiovascular, musculoskeletal, hepatobiliary.

Tissue/cell level. Intestinal epithelium (M cells), lymphoid tissue (Peyer's patches, UBERON:0001211), vascular endothelium (coronary artery endothelial cells), synovium. Cell populations: CD4⁺/CD8⁺ T cells (Vβ3/9/13⁺), professional phagocytes, epithelioid histiocytes forming granulomas with giant cells.

Subcellular level. T-cell plasma membrane (TCR–MHC-II synapse); cytokine secretory machinery (ER/Golgi). Suggested GO cellular-component terms: GO:0009897 (external side of plasma membrane), GO:0042101 (T cell receptor complex), GO:0042613 (MHC class II protein complex).

Localization/lateralization. Terminal ileitis is typically right-lower-quadrant, mimicking appendicitis; coronary lesions may be bilateral; reactive arthritis is often asymmetric oligoarticular (lower limbs) but can be polyarticular ([PMID: 12922960](#)).

8. Temporal Development

Onset. Acute, pediatric-predominant; incubation typically a few days to ~2 weeks after ingestion. In the case series the illness was acute and multisystem ([PMID: 6344044](#)).

Progression and course. Most acute illness is **self-limited** over 1–3 weeks. A minority progress to systemic toxicity (toxic-shock-like), DIC (PMID: 16366361), coronary involvement, or septicemia (higher risk in immunocompromised/iron-overloaded hosts, PMID: 42448289). Post-infectious reactive arthritis appears days to weeks after the acute phase and may persist >6 months in a substantial fraction; a 10-year follow-up found chronic joint symptoms in 9/16 patients including ankylosing spondylitis and fatal secondary amyloidosis (PMID: 7554560).

Patterns. Acute phase resolves spontaneously or with antibiotics; complications may be relapsing (reactive arthritis reactivation) or progressive (spondyloarthropathy/amyloidosis). The critical intervention window for KD-like cases is the first ~10 days (IVIG to prevent coronary aneurysm).

9. Inheritance and Population

Inheritance. None — FESLF is infectious, not heritable. No Mendelian inheritance pattern, penetrance, expressivity, anticipation, mosaicism, founder effect, consanguinity, or carrier frequency applies to the disease itself. (Host modifiers such as HLA-B27 follow their own inheritance but only affect complication risk.)

Epidemiology and geography. FESLF is **geographically restricted** to the Russian Far East, Siberia, and Japan, occurring sporadically and in outbreaks (PMID: 26819960; PMID: 32498317). This restriction tracks the geographic distribution of ypmA⁺ Asian-clade strains: ypmA detected in 96.2% of 212 Siberian/Far-Eastern strains (PMID: 17163133) and 88.9% of fatal Japanese monkey-outbreak strains (PMID: 18242014). By contrast, European *Y. pseudotuberculosis* infection is sporadic self-limiting gastroenteritis (PMID: 26819960). General yersiniosis incidence in non-endemic regions is low (e.g., ~0.16/100,000/yr in a multistate US study; *Y. pseudotuberculosis* is a small minority of cases) (PMID: 25931631; PMID: 26233079).

Demographics. Children are predominantly affected; Yersinia-associated KD onset is older than Yersinia-negative KD (3.05 vs 2.31 y; p = 0.03) (PMID: 17129979). Sex ratio is not strongly skewed for the acute disease; HLA-B27-associated reactive arthritis affects both sexes.

Population structure of the pathogen. MLST reveals a worldwide **cluster A** (ypmA, pYV) and a Far-East-restricted **cluster B** (ypmB); the ypm superantigen genes are distributed across the phylogeny with ypmA in cluster A and ypmB in cluster B (PMID: 21131531). FESLF strains belong to the **Asian clade** and are KD-related (PMID: 39780644).

10. Diagnostics

Microbiology. Stool culture and/or blood culture for *Y. pseudotuberculosis* (cold enrichment improves yield); serotyping (O:1, O:3, 4b). Blood culture positivity in septicemic cases ([PMID: 18411766](#)).

Serology. ≥ 4 -fold rise in agglutinating antibody titers is diagnostic ([PMID: 6344044](#)); tube agglutination titers (e.g., $\geq 1:160$ against serotype 4b) support diagnosis ([PMID: 18411766](#)). **Anti-YPM (anti-mitogen) antibody titers** rise between days 7–18 and confirm infection when cultures are negative ([PMID: 34108299](#); [PMID: 16366361](#)). Serology is described as the most informative laboratory approach ([PMID: 22416431](#)).

Molecular. PCR for virulence genes — ypm (ypmA/B/C), virF, inv, irp2 — enables genotype-based confirmation and epidemiologic typing ([PMID: 18242014](#); [PMID: 22416431](#)).

Laboratory abnormalities. Leukocytosis, elevated CRP/ferritin, transaminase elevation (hepatic involvement), thrombocytopenia and prolonged PT/PTT with elevated FDP in DIC ([PMID: 16366361](#)).

Imaging/histopathology. Ultrasound/CT show mesenteric lymphadenopathy and terminal ileitis; echocardiography detects coronary dilation/aneurysm in KD-like cases. **Biopsy** shows epithelioid granulomas with reticular microabscesses/stellate necrosis and giant cells, mimicking Crohn's disease ([PMID: 22228001](#); [PMID: 18368812](#); [PMID: 26385573](#)).

Clinical criteria / differential diagnosis. When patients meet Kawasaki-disease criteria (fever ≥ 5 days plus ≥ 4 of: rash, conjunctivitis, oral changes, extremity changes, cervical lymphadenopathy), *Y. pseudotuberculosis* should be excluded ([PMID: 39697956](#)). Differentials: scarlet fever (Group A *Streptococcus*), Kawasaki disease, appendicitis, Crohn's disease, mesenteric adenitis of other cause, other yersiniosis. Genetic testing is **not applicable** (no germline cause).

11. Outcome / Prognosis

Survival/mortality. The acute disease is usually self-limited with low mortality when treated. Severe outcomes include septicemia with multi-organ dysfunction (fatal cases reported, especially in immunocompromised/iron-overload hosts, [PMID: 42448289](#)) and DIC ([PMID: 16366361](#)). Long-term fatal outcomes are rare and usually via reactive-arthritis complications (secondary amyloidosis $\rightarrow$ uremia) ([PMID: 7554560](#)).

Morbidity/complications. Coronary artery lesions (KD phenotype): significantly more frequent in *Yersinia*-positive KD (22/42, 52.4%) than *Yersinia*-negative KD (105/330, 31.8%; $p = 0.001$), with greater need for additional IVIG (36.1% vs 16.0%; $p = 0.004$) (PMID: 17129979). Post-infectious reactive arthritis in 12–22% of adults, HLA-B27-associated, sometimes chronic/polyarticular (PMID: 23852698; PMID: 12922960); erythema nodosum (42% of children in one outbreak); rarely ankylosing spondylitis and amyloidosis.

Prognostic factors. Presence of *ypmA*⁺ Far-Eastern strain and pVM82 plasmid (severity); host HLA-B27 (reactive-arthritis risk); older age and *Yersinia* positivity (coronary-lesion risk and IVIG resistance in KD); immunocompromise/iron overload (septicemia risk). Anti-YPM antibody serology serves as a diagnostic/prognostic biomarker.

12. Treatment

Pharmacotherapy (antibiotics). *Y. pseudotuberculosis* is generally susceptible to third-generation cephalosporins (cefotaxime, ceftriaxone), fluoroquinolones, aminoglycosides, tetracyclines, and trimethoprim-sulfamethoxazole; carbapenems (imipenem) used in severe/septicemic disease (PMID: 18411766). Refractory KD-presenting cases have responded to cefotaxime after immunosuppressive therapy (PMID: 39697956). Suggested NCIT: C264 (Cephalosporin), C540 (Ciprofloxacin), C61796 (Cefotaxime).

Kawasaki-disease-presenting cases. Intravenous immunoglobulin (IVIG) plus aspirin to reduce coronary complications; note higher IVIG-resistance in *Yersinia*-associated KD, sometimes requiring additional IVIG or immunosuppression (PMID: 17129979; PMID: 39697956). NCIT: C488 (Immunoglobulin therapy), C287 (Aspirin).

Reactive arthritis. NSAIDs, and in chronic/severe spondyloarthropathy, DMARDs; supportive rheumatologic care.

Supportive care. Fluid/electrolyte management for diarrhea/vomiting; DIC management; hepatic monitoring.

Advanced/experimental therapeutics. A conceptual therapeutic avenue arises from mutagenesis work: engineered YPM point mutants with reduced TCR-V β binding lose overstimulatory/cytotoxic activity and could serve as immunotherapeutic/vaccine antigens (PMID: 10087177). Anti-IFN- γ and anti-YPM antibodies prevented liver injury and death in mice — a proof-of-concept for toxin/cytokine-neutralizing therapy (PMID: 15003813). No approved gene, cell, or RNA therapy exists or is applicable.

Pharmacogenomics. Not established for FESLF.

13. Prevention

Primary prevention. Food and water hygiene: thorough washing of raw vegetables (carrots, leafy greens), safe water, cold-chain awareness (organism grows at refrigeration temperatures). Outbreak control through tracing contaminated produce ([PMID: 23852698](#)).

Immunization. No licensed human vaccine against *Y. pseudotuberculosis*/FESLF. An attenuated *Y. pseudotuberculosis* strain (lacking HPI, ypm, pil; retaining pYV) has been used experimentally as an oral live vaccine against plague, protecting 75–88% of mice — illustrating vaccine-platform potential but not a FESLF vaccine ([PMID: 18505804](#)).

Secondary/tertiary prevention. Early antibiotic treatment; early echocardiography and IVIG in KD-presenting cases to prevent coronary aneurysm; monitoring/treatment of reactive arthritis to prevent chronic sequelae.

Public health. Surveillance of foodborne yersiniosis; produce-supply monitoring; reservoir awareness (rodents, wild boars, birds). Vector control is not relevant (foodborne, not vector-borne).

14. Other Species / Natural Disease

Taxonomy of causative agent. *Yersinia pseudotuberculosis* (NCBI:txid633).

Natural disease in animals. Pseudotuberculosis is a common cause of mortality in captive exotic birds and mammals; per OIE WAHIS-Wild it is the 8th most frequently reported disease/infection in wildlife worldwide and 5th in wild mammals ([PMID: 42431144](#)). Fatal outbreaks in **breeding monkeys** in Japanese zoos (28 deaths, 8 species; 88.9% ypmA⁺) directly parallel human FESLF and implicate YPM in high mortality ([PMID: 18242014](#)). Documented in wild boars ([PMID: 29980552](#)) and, newly, free-ranging marine odontocetes ([PMID: 42431144](#)).

Zoonotic potential / reservoirs. Rodents, birds, and other mammals serve as reservoirs; transmission to humans is principally by ingestion of contaminated food or water. Cross-species susceptibility is broad.

Comparative biology. The superantigen mechanism is conserved: ypmA⁺ strains cause severe/fatal systemic disease across primates and humans, supporting evolutionary conservation of the YPM–MHC-II/TCR-V β axis.

15. Model Organisms

Mouse models. BALB/c mice develop YPM-induced toxic shock; **T-cell-deficient SCID mice do not**, establishing T-cell dependence (PMID: 15003813). Mini-osmotic-pump delivery of YPM produces protracted V β 3⁺CD4⁺ T-cell expansion and immunological memory, modeling chronic superantigen exposure (PMID: 11937534). A murine coronary-arteritis (KD) model is induced by oral microbe-associated molecular patterns, linking *Yersinia* biofilm MAMPs to KD-like vasculitis (PMID: 25411968).

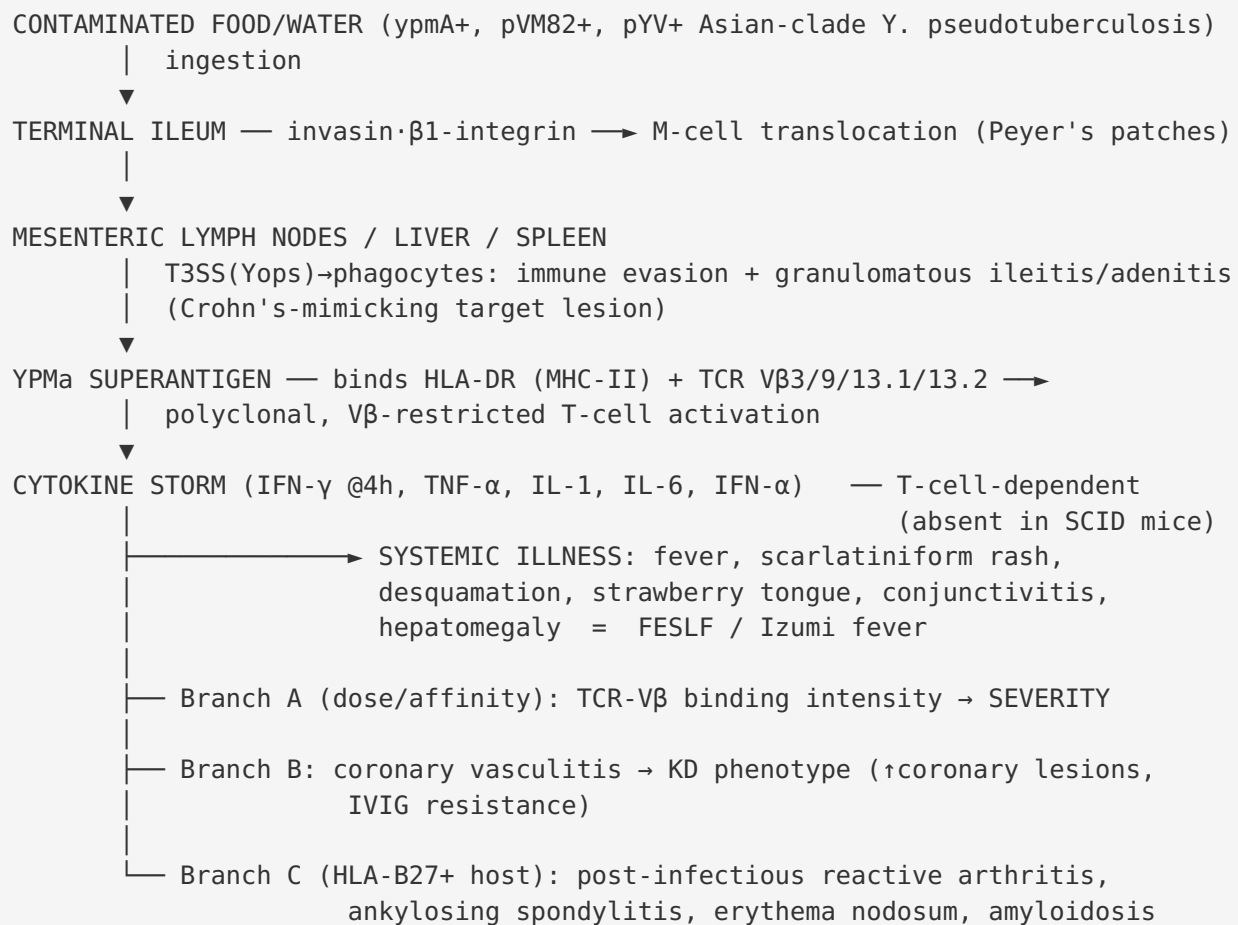
Rat models. HLA-B27 transgenic rats mount a CD8⁺ CTL response to *Y. pseudotuberculosis*; HLA-B27 exerts a negative effect on this response, modeling impaired defense and the HLA-B27–reactive-arthritis link (PMID: 10417137). LEW rat BV8S2⁺ T cells respond to YPM, used to map TCR CDR/HV4 contributions to superantigen recognition (PMID: 15096488).

Non-human primates. Naturally infected breeding monkeys constitute a spontaneous high-fidelity model of fatal systemic ypmA⁺ disease (PMID: 18242014).

In vitro / structural. Human whole-blood cytokine assays (YPM elicits maximal IL-1/IL-6/IFN- α /TNF- α , PMID: 12449699); recombinant YPM mutagenesis systems for structure–function (PMID: 10406939; PMID: 10087177); X-ray crystallography/NMR of YPMa (PMID: 12832802; PMID: 17369701).

Recapitulation/limitations. Mouse models capture superantigen-driven T-cell activation and toxic shock but incompletely reproduce the full mucosal→systemic human sequence; the HLA-B27 rat captures the reactive-arthritis modifier but not the acute scarlatiniform phenotype.

Mechanistic Model / Interpretation



The unifying interpretation is that FESLF is fundamentally a **superantigen toxicosis superimposed on an enteric invasive infection**. What distinguishes the Far-Eastern severe disease from European mild yersiniosis is not the route or the organism per se but the horizontally acquired **ypmA** superantigen (plus the enigmatic **pVM82** plasmid). Severity is a graded function of TCR- $V\beta$ engagement, and the host HLA background (HLA-DR for presentation, HLA-B27 for post-infectious autoimmunity) shapes both acute magnitude and chronic sequelae. This model explains the disease's geographic restriction, its overlap with scarlet fever and Kawasaki disease, and the rationale for both antibiotic (source control) and immunomodulatory (IVIG/anti-cytokine) treatment.

Evidence Base

PMID	Contribution
6344044	Defines the multisystem scarlet-fever/KD-like phenotype; links to Izumi fever (12-child series)
39780644	FESLF strains = Asian clade, KD-related (genomics)
26819960	Review: FESLF definition, Russia/Japan geography, YPMa role
30695393	pVM82 plasmid unique to FESLF clinical-epidemic strains
17163133	ypmA in 96.2% of Far-Eastern strains; pYV ⁺ /ypmA ⁺ /HPI ⁻ systemic genogroup
18242014	88.9% ypmA ⁺ in fatal Japanese monkey outbreaks
15784605	Horizontal acquisition/linkage of ypm and pil
21131531	Population structure: cluster A (ypmA) worldwide, cluster B (ypmB) Far-East
15003813	T-cell-dependent YPM toxicity; V β 8 ⁺ migration to liver; IFN- γ surge; antibody protection
12449699	YPM = maximal cytokine inducer among Yersinia stimuli
12832802	YPM crystallization; no homology to other superantigens
17369701	Jelly-roll fold of YPMa; YPMa/b/c family
10406939	YPM competes with SEE for HLA-DR; MHC-II/TCR mapping; essential S-S bond
9287138	YPMb shares V β 3/9/13.1/13.2 specificity
10087177	TCR-V β binding intensity determines pathogenic severity
17129979	Yersinia-KD: $\uparrow$ coronary lesions (52.4% vs 31.8%, p=0.001), IVIG resistance
34108299	Anti-YPM antibody serology confirms infection; intussusception + incomplete KD
16366361	KD-criteria case with DIC; anti-YPM serology
39697956	Refractory KD with Y. pseudotuberculosis treated with cefotaxime
10592892	Superantigen hypothesis for KD pathogenesis
23852698	O:1 carrot outbreak: ReA 22% adults, 67% HLA-B27 ⁺ , EN 42% children
12922960	O:3 outbreak: severe polyarticular ReA, HLA-B27 ⁺
7554560	10-yr follow-up: chronic arthritis, ankylosing spondylitis, fatal amyloidosis

PMID	Contribution
34628159	Meta-analysis: <i>Yersinia</i> 65% terminal ileitis, 51% mesenteric adenitis
22228001	Terminal ileitis mimicking Crohn's in childhood
18368812	Granulomatous <i>Yersinia</i> pathology resembling Crohn's
26385573	Histopathologic IBD mimics incl. <i>Yersinia</i>
25576025	Invasin- β 1-integrin $\rightarrow$ NETs (innate arm)
20148898	T3SS Yop translocation selectively to phagocytes
18505804	Attenuated <i>Y. pseudotuberculosis</i> oral vaccine concept
10417137	HLA-B27 transgenic rat CTL model
42448289	Fatal septicemia in immunocompromised/iron-overload host

Limitations and Knowledge Gaps

- No dedicated primary data were analyzed** — this is a literature synthesis. Some key sources are non-English (Russian/Japanese) with abstracts unavailable, limiting extraction of quantitative detail (e.g., precise FESLF incidence/prevalence per 100,000, which remains unquantified in the accessible literature).
- pVM82 plasmid biology is understudied** — it is a molecular marker of FESLF strains, but its gene content and mechanistic contribution to severity are essentially uncharacterized ([PMID: 30695393](#)).
- Direct in vivo demonstration that YPMa alone reproduces the full FESLF rash/desquamation phenotype is lacking** — most mechanistic data derive from toxic-shock and cytokine models rather than the complete syndrome.
- Causality vs. association in Kawasaki disease** — the *Yersinia*–KD link is strong epidemiologically and serologically, but whether *Y. pseudotuberculosis* is a cause of a KD subset or a mimicker remains debated.
- Epidemiologic underreporting** — yersiniosis is under-diagnosed and under-serotyped even in surveillance systems ([PMID: 26233079](#)), so true FESLF burden in endemic regions is uncertain.

6. **No modern randomized treatment trials** — antibiotic and IVIG recommendations rest on case series and analogy to KD, not RCTs.

Proposed Follow-up Experiments / Actions

1. **Sequence and functionally annotate the pVM82 plasmid** across FESLF vs non-FESLF strains (comparative genomics + isogenic cured-plasmid virulence assays) to test whether pVM82 independently contributes to severity beyond ypmA.
 2. **Isogenic Δ ypmA vs ypmA⁺ challenge** in a humanized-HLA-DR/V β mouse or primate model to formally establish YPMa's necessity/sufficiency for the FESLF phenotype.
 3. **Prospective serologic cohort** of KD patients in endemic vs non-endemic regions using standardized anti-YPM assays to quantify the attributable fraction of Yersinia-associated KD and its coronary-outcome risk.
 4. **Structure-guided anti-YPM therapeutics**: develop neutralizing monoclonal antibodies or engineered V β -decoy proteins based on the jelly-roll fold and the demonstrated protection by anti-YPM/anti-IFN- γ antibodies ([PMID: 15003813](#)).
 5. **HLA-B27 stratified natural-history study** to define reactive-arthritis and amyloidosis risk after *Y. pseudotuberculosis* infection and guide surveillance.
 6. **Regional epidemiologic quantification** in the Russian Far East/Japan to populate incidence/prevalence fields (currently unfilled) and map ypmA/pVM82 genotype distribution to case severity.
 7. **Update ontology mappings** for the knowledge base: confirm MONDO:0041536 cross-references to MeSH/ICD, and attach the HPO, GO, CL, UBERON, and NCIT terms proposed in Sections 3–12.
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Report compiled from 65 reviewed papers and 8 confirmed findings over a 5-iteration autonomous investigation. Evidence types: human clinical (case series, outbreaks, cohorts), model organism (mouse, rat, primate), in vitro (cytokine assays, mutagenesis), and structural/computational (X-ray crystallography, NMR, genomics).
